

Question 1: Simple Calculator using `switch` statement

Write a C++ program that acts as a simple calculator. The program should:

1. Prompt the user to enter two floating-point numbers.
2. Prompt the user to choose an operation: `+` (addition), `-` (subtraction), `\*` (multiplication), or `/` (division).
3. Use a `switch` statement to perform the chosen operation on the two numbers.
4. Display the result.
5. Handle the case of division by zero (e.g., print an error message) using an `if-else` statement *within* the division case.
6. Include a `default` case in the `switch` statement to handle invalid operator input.

****Example Interaction:****

...

Enter first number: 10.5

Enter second number: 2.0

Choose operation (+, -, *, /): *

Result: 21.0

...

Question 2: Grade Calculator and Day of the Week Identifier

This question requires you to implement two distinct functionalities, demonstrating the appropriate use of `if-else if-else` and `switch` statements.

Part A: Grade Calculator (using `if-else if-else`)

Write a C++ program that prompts the user to enter a numeric score (0-100). The program should then use `if-else if-else` statements to determine and print the corresponding letter grade based on the following scale:

- * 90-100: A
- * 80-89: B
- * 70-79: C
- * 60-69: D
- * 0-59: F
- * Handle invalid scores (outside 0-100 range) by printing an error message.

Part B: Day of the Week Identifier (using `switch`)

Write a C++ program that prompts the user to enter an integer from 1 to 7. Use a `switch` statement to display the corresponding day of the week (e.g., 1 for Monday, 2 for Tuesday, ..., 7 for Sunday). Include a `default` case to handle invalid input (numbers outside 1-7).

Question 3: Character Type Analyzer

Write a C++ program that prompts the user to enter a single character. The program should then analyze the character and determine if it is:

1. A vowel (case-insensitive: 'a', 'e', 'i', 'o', 'u', 'A', 'E', 'I', 'O', 'U').
2. A consonant (any other English alphabet letter).
3. Neither a vowel nor a consonant (i.e., not an alphabet character).

You **must** use a `switch` statement to identify vowels. For checking if a character is an alphabet and then determining if it's a consonant, you should use `if-else` statements.

Example Interactions:

...

Enter a character: a

'a' is a vowel.

Enter a character: B

'B' is a consonant.

Enter a character: 7

'7' is not an alphabet character.

...